FETCH

What each part of the system does, and how its forecast works

Inventory Management System with Predictive Analytics for Risha Pet Supplies

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Written in plain words, for explaining the system to someone who has not used it.

Part 1 goes through each function of the system: what it is for, what you can do there, and what the system checks for you.

Part 2 explains the analytics step by step: what the forecast reads, how it is worked out, how its accuracy was measured, and what it cannot do. The figures are the real ones.

11 October 2026. This describes the system on the work branch, after the fixes that followed the QA test. The live site does not have these changes until they are deployed.

Every figure in Part 2 comes from running the system's own forecasting code on the shop's sales. The technical version, with the commands to repeat each check, is in QA/FORECAST-METHOD.md.

Part 1. What each part of the system does

1. The system in one paragraph

FETCH keeps one record of everything the shop has on the shelf. Every sale at the till takes stock off that record at once, so the back office always sees what is really left. From the sales it has recorded, the system estimates what each product will sell in the next 30 days, and turns that into three answers: what to reorder, how much, and by which date. It runs in a web browser. Nothing is installed on the shop's computers.

2. Who uses it

Each person signs in with their own account. The account has a role, and the role decides which screens open and which buttons work.

Role	What they use	What they cannot do
Administrator	Everything in the back office: products, suppliers, reports, users, the activity log, email settings and backups	Does not use the till
Manager	Products, stock, suppliers, reorders, voiding a sale	Cannot manage user accounts. Does not use the till
Staff (cashier)	The till only: sell, record cash put in or taken out, see their own sales today, count the drawer at the end of the day, ask for a sale to be voided	Cannot open the back office
Viewer	Can look at everything in the back office	Cannot change anything
Supplier	An outside login tied to one supplier. Sees only the orders sent to them and the products they supply. Can confirm an order and mark it shipped	Cannot see the shop's other suppliers, sales or stock

After signing in, each person is taken to their own screen. The role is checked again by the server on every request. Hiding a button is a convenience; the server's check is what keeps someone out.

3. Signing in and accounts

What it is for

Making sure every action in the system belongs to a known person.

What you can do

- Sign in with a user name and a password.
- Tick **Keep me signed in** to stay signed in on that browser for 7 days. Without it, the session ends when the browser is closed, and after 24 hours at most. Leave it off on a shared till.
- **Forgot your password?** An administrator is emailed a reset link straight away. Anyone else waits for an administrator to approve the request, and is then emailed the link.

- A new account confirms its email address through a link before it can sign in.

What the system checks for you

- Passwords need at least 6 characters, with a capital letter, a number and a special character.
- Five refused sign-ins from one place lock sign-in from that place for 15 minutes.
- A wrong password and a user name that does not exist get the same answer and take the same time, so nobody can find out which names exist by trying them.
- A deactivated account cannot sign in, and a session it had open stops at once.

4. Dashboard

What it is for

The state of the shop today, on one screen. It is the first screen an administrator, manager or viewer sees.

What it shows

- Headline figures: number of products, revenue, number of sales, and how many products need restocking. Voided sales are not counted as sales; the card says how many more were voided.
- A chart of the last 30 days of sales.
- Recent sales, each with its own status (Completed or Voided).
- Products that need restocking, products close to their expiry date, and purchase orders still open.
- A **Live** marker, which turns red if the connection drops.

5. Inventory

What it is for

The list of everything the shop sells, and the stock of each product.

Finding products

- View as cards or as a table. Search by name, SKU, brand or barcode.
- Filter by status: Low stock, Out of stock, Expiring soon, Expired, Healthy. Filter by category or by animal.
- The cards at the top read: Total products, Restock needed, Expiring soon, Total inventory value, Over max level.

Adding and changing a product

- **Add product.** Five things are required: SKU, name, category, unit price and supplier. A product without them could not be found, sold or reordered.
- **Edit** any field later. **Delete** hides the product: it leaves Inventory, the till and the reorder lists, while its past sales stay in the reports.

- **Import** many products at once from a spreadsheet file (a template is provided). Rows with a problem are left out and named, with the reason.
- Barcode and picture tools add a barcode or a photo to products that have none.

Changing stock by hand

The **Stock** button opens a small window with four choices. Each says which way it goes:

Choice	Effect on stock
Add stock (received, found, returned)	Goes up
Remove stock (taken out)	Goes down
Remove stock: count was lower than the system	Goes down
Remove stock: damaged or spoiled	Goes down

The window shows the stock before and after. Every change is recorded with who made it and why.

Reordering

Reorder lets you pick low-stock products. After a final confirmation it creates the purchase orders and emails each one to that product's supplier.

What the system checks for you

- A missing required field is named. A SKU that is already in use is refused.
- A name that another product already has is asked about first ("Save this one under the same name anyway?"), so two sizes can share a name but not by accident.
- More stock cannot be removed than there is ("Only 20 in stock: you cannot remove 999999.").
- An expiration date cannot be in the past. A number too large, or text too long, is refused with the field named.
- Pressing Save twice saves once.
- Deleting asks first, names the product, and says so if it still has stock on the shelf.

6. Point of sale (the till)

What it is for

Selling. Only staff accounts use it, and it is the only screen they have.

What you can do

- Search or browse products on the left and click to add them to the cart. Scan a barcode at any time (F2 puts the cursor in the scan box).
- On the right: change quantities, give a discount, type the cash received (quick buttons: Exact, +20, +50, +100). The change is worked out for you.
- **Charge** saves the sale, takes the items off the stock at once, and shows a receipt that can be printed.

- Record the starting cash in the drawer, and cash put in or taken out during the day.
- See your own sales for today, and ask for one to be voided, giving the reason. A manager decides.
- Count the drawer at the end of the day. The cashier types what they counted before the system shows what it expected, so the answer cannot be read off the screen.

What the system checks for you

- A product that is out of stock or expired cannot be sold.
- A sale cannot take more than the stock, even if the same product appears on two lines.
- The price is the system's own: a price changed on the page is ignored.
- A discount above the limit is refused. The limit is 10% unless the administrator changes it.
- Pressing Charge twice, or a slow connection, cannot save the same sale twice.
- Payment is cash only.

7. Sales history

What it is for

Every sale ever made, for looking up and for correcting mistakes.

What you can do

- Search, and filter by date. Open a sale to see its items, who sold it, its status, and a printable receipt.
- **Void** (administrator or manager): cancels a sale, with a required reason. The items go back into stock. The sale stays in the history marked Voided and is left out of every total.
- **Delete** (administrator only): removes a sale. The activity log keeps a copy of it: its number, date, cashier, customer, totals and every item.
- **Export** saves the list as a spreadsheet file, named with the shop's date.
- **End of day**: the system shows the cash expected from the day's sales, you enter what was counted, and any difference is saved with a note.

What the system checks for you

- A start date after the end date is refused with the reason, as is a date that does not exist.
- A voided sale's items are returned to stock once, never twice.

8. Suppliers and purchase orders

What it is for

Who the shop buys from, and everything on order.

What you can do

- Keep each supplier's contacts and payment terms, and see how well they deliver.

- See every email sent to a supplier, with a mark showing whether it was delivered and whether they opened it.
- The **Orders** view shows orders on the way: the date the supplier promised, any item they cannot fully supply, and their delivery receipt number. A late order is flagged.
- When goods arrive, **Receive into stock** adds them to the stock and can set the new batch's expiration date.
- The same view lists offers a supplier has suggested, price changes waiting for a decision, and deliveries not yet paid.

The supplier's own page

A supplier with a login sees only their own part: open orders, past orders, the products they supply, payments, how they are doing, and their details on file. They confirm an order and mark it shipped themselves.

What the system checks for you

- Two suppliers cannot have the same name. An email address or phone number has to look like one.
- A product needs a current supplier before it can be reordered.

9. Reports

Four tabs.

Tab	What it shows
Sales	Pick a period (last 7 days, last 30 days, this month, all time, or your own dates) and view it by day, week or month. Revenue, number of sales, average per sale, items sold, best-selling products and sales by category. The category list adds up to revenue: sales recorded without items, and discounts, are shown as their own lines. Export saves the table.
Stock planning	The forecast, and what to do about it. Part 2 of this document explains it.
Void requests	Sales a cashier has asked to have voided, with the reason. Choose Void sale or Keep sale. Recently voided sales are listed below. Administrators and managers only.
Activity log	Who did what, and when. Administrators only.

10. Notifications

Notices the system raises by itself: a product running low, a product that has run out, a product close to its expiry date, a product over its maximum level, a purchase order changing status, requests to reset a password, and routine events such as a stock change. They open from the bell in the side menu or from the Notifications page.

- Suppliers of low-stock products are emailed automatically, at most once a day.
- The list is shared. Only an administrator or a manager can mark notices read or clear them; a viewer can read them.

11. Users and roles

For the administrator only.

- Add an account and give it a role. Edit it later.
- **Deactivate** an account: it can no longer sign in, and nothing it recorded is removed. Its user name and email stay reserved. **Restore** brings it back.
- Send the email confirmation link again.
- Approve or decline a request to reset a password.

The system will not let the last administrator be removed, and nobody can remove their own account or change their own role.

12. Settings

Section	What is there
Profile information	Your name, email, phone and picture
Change password	Set a new password
Preferences	Appearance (the look of the screens). For the administrator: the largest discount allowed at the till, and Data backup, which downloads a copy of the whole database as a zip file
Email configuration	For the administrator: the address the system sends from, and a button to send a test email
Session	Your current sign-in: your role, user name and status

13. The assistant

A chat helper on the back-office pages. You type a question about the shop or about how to use the system, and it answers in plain words.

How it works: the system gathers a short summary of the shop as it is now (how many products, what needs restocking, what is expiring) and sends it with your question to Google's Gemini service, which writes the answer. It is told to answer only about this system. It needs a key to be set up, and it sends that summary outside the shop each time a question is asked.

14. What keeps the records safe

Protection	What it does
Sign-in on everything	No page and no data can be opened without signing in
Roles checked by the server	A viewer or cashier cannot do an administrator's action, even with a hand-made request
One press, one save	A double-click or a slow connection cannot save the same sale, stock change, product or supplier twice

Protection	What it does
Stock cannot go below zero	Neither a sale nor a stock change can take more than there is
Ask before anything is removed	Deleting, voiding and deactivating each ask first and say what will happen
Activity log	Every add, change, delete, void and sign-in is recorded with who and when
Backups	A copy of the whole database is emailed to the shop once a day, and the administrator can download one at any time
Input checks	Missing fields, impossible dates, oversized numbers and text typed where a number belongs are refused with a clear message
Typed text stays text	Anything typed into a field is shown as plain text and never run as code

Part 2. The analytics: how the forecast works

1. The question it answers

How many of each product will the shop sell on each of the next 30 days?

From that one answer, the Stock planning screen works out the rest: how long the stock will last, when it will run out, the last day to place an order, how much to order, which products have too much stock, and which will expire before they sell.

The method in one sentence: two machine-learning models, **Random Forest** and **Gradient Boosting**, learn from all products' sales together and their two answers are averaged.

2. Why forecasting this shop is hard

It helps to know this before reading any accuracy figure.

- The shop has 120 products. At present it sells about 11 items a day across all of them.
- The busiest product sold 19 in the last 8 weeks: about 1 every 3 days.
- No product sold on more than 13 of the last 56 days. On 9 product-days out of 10, nothing is sold.

When a product sells one or two items in a fortnight, nobody can say which day the next one will sell. So a forecast for a single product is always rough. Added up over a category, or the whole shop, the forecast is much steadier. **The forecast is most useful for totals.**

3. What it reads

Which sales	Completed sales only. A voided sale is not counted.
How far back	Up to 12 months. The shop's records start on 28 February 2026, so at present that is 225 days: 2,338 sales, 10,370 items.
What a "day" is	The shop's own calendar day (Philippine time).
Today	Left out. The day is not over, so counting it would treat a morning's sales as a whole day's.
A day when a product did not sell	Counted as zero for that product.
How often it is worked out	When someone opens Stock planning, at most once every 5 minutes.

4. First, three safeguards

The models learn from whatever they are shown. Three things they must not learn from are dealt with first.

Safeguard 1: days without data are left out

A day when nothing sold is normally a real zero, and the forecast should learn from it. But two kinds of empty day say nothing about what customers want:

- days on which sales were recorded but none of them has its items
- three or more days in a row with no sale at all (the shop was closed, or nothing was being recorded)

In the shop's records there is one such stretch: **10 to 31 May 2026, 22 days**. From 10 to 27 May there are 199 sales (#1097 to #1295, ₱1,259,362.00) recorded with no items, and 28 to 31 May have no sales at all.

Those 22 days are skipped when the history is read. The forecast does not learn "nothing sells" from them. **Nothing in the database is changed**, and the 199 sales are untouched. Stock planning says which days were not counted, and Reports show the ₱1,259,362.00 as its own line so that revenue by category adds up:

```
categories ₱8,847,885.00 + sales with no items ₱1,259,362.00 − discounts ₱123,724.50 = revenue ₱9,983,522.50
```

Safeguard 2: a freak day is capped

One model is fitted for all products together. So one freak day in one product (a mistyped quantity, a single bulk purchase) used to bend the forecast of every product.

Now, before anything is learned, one day counts for no more than a cap:

```
cap = the larger of 5, and 4 × what the product sells on a typical selling day
```

"Typical" means the amount the product reaches or stays under on three of every four days it sells. A product with fewer than 5 selling days is judged by the same figure for the whole shop. On the shop's data the cap touches 6 of 3,971 selling days.

Made-up test: a product added to the 120 real ones	Effect on the other products, before	Now
Ordinary: 1 a day	−3.2%	+3.8%
The same, with one day of 10,000	+130%	+4.1%
The same, with one day of 1,000,000	+15,044%	+4.1%
The same, with 300 sold on the last day	+22.8%	+4.1%

So a freak day now adds almost nothing beyond what the same product would do anyway.

Safeguard 3: a new product gets a cautious first estimate

A daily rate cannot be read from a handful of days. A product that sold 5 on its first day used to be forecast at 5 every day: 150 in 30 days, with an order suggested for 165.

Until a product has 14 days on record, what it has sold so far is spread over 14 days:

```
daily rate = items sold so far ÷ 14
```

Made-up test	Before	Now
One sale of 1, yesterday	30 expected in 30 days	2.1
One sale of 5, yesterday	150 expected, "reorder now, order 165"	10.8, no order suggested

The page says so on such a product: "…this is a cautious first estimate". The estimate grows as more is sold, and the normal forecast takes over on day 14.

5. Which method forecasts which product

Days of sales on record, from its first sale	Method	Why
Under 14	Cautious first estimate (above)	Too little to read a rate from
14 to 41	Simple trend	Enough to see a level, too little for the models
42 or more	The two machine-learning models	The models need 28 days to build their inputs and 14 to learn from

If the machine-learning program cannot run for any reason, every product falls back to the simple trend, and the page says so. The forecast never simply fails.

6. The two machine-learning models

The idea of a decision tree

A decision tree is a chain of yes-or-no questions that ends in a number. For example: "Is it a weekend? Did this product sell anything last week? Is its average over the last month above 0.3 a day?" Each path of answers ends in an estimate. The tree works out by itself, from past sales, which questions to ask.

Random Forest

One tree can be misled by one odd day. A Random Forest grows **60 trees**, each on a different random sample of the past, and takes the average of their answers. An odd day misleads a few trees, not most of them.

Gradient Boosting

Gradient Boosting builds **80 small trees one after another**. Each new tree looks only at the mistakes the earlier ones left, and corrects a little of them. It picks up patterns the forest smooths over.

The two together

```
forecast for a day = (Random Forest + Gradient Boosting) ÷ 2, never below zero
```

Both come from scikit-learn, a standard machine-learning library for Python. The same sales always give the same forecast: nothing in it is left to chance from one run to the next.

One model for the whole shop

The two models are fitted once on every product's history together, not once per product. A single product here has a few dozen days on which anything sold. Learning from it alone, a tree would memorise the odd day. Together there are thousands of product-days to learn what a weekend, a pay-day week or a quiet month does to sales.

The 12 things the model looks at

About	What it looks at
The day being forecast	Day of the week Weekend or not Day of the month Month Public holiday or not (14 fixed-date Philippine holidays)
The product's recent sales	Sales on the last day on record Sales a week before the first day forecast Sales two weeks before it Sales four weeks before it
The product's level	Average of the last 7 days Average of the last 28 days
How far ahead	How many days ahead the forecast is (1 to 30)

Nothing tells the model which product it is looking at. What tells it how much a product sells is that product's own recent sales and averages.

How much history it learns from

- The model learns from the **last 120 days** on record, about four months.
- **Recent days count for more.** A day's weight halves every 60 days going back.

The reason: the shop's sales fell to about a quarter of their earlier level in June (100 to 135 items a day in March to May, 26 in June, 11 in October). A model that treated the old level and the new one as equals read the present too high.

Each day is forecast directly

The usual shortcut is to forecast tomorrow, pretend that happened, and forecast the day after from it. For a shop where most products sell on few days, that drifts upward. Here each of the 30 days is forecast straight from the last real sales, and nothing the model predicts is fed back in.

7. The simple trend

Used for a product with 14 to 41 days on record. It reads the product's last 8 weeks and mixes three ordinary methods:

Part	Share	In plain words
Holt's smoothing	50%	A running average that follows the current level and the direction it is moving in
Linear regression	30%	A straight line drawn through the 8 weeks, carried forward at most 14 days
Weighted moving average	20%	The last 7 days, the most recent counting most

- For a slow product (under 1 a day), most of the figure is simply its average over the last 28 days. Trend methods chase noise on a product that sells now and then.
- The figure never falls below 55% of the product's average over its last 14 days.
- Each day is then nudged up or down by how that day of the week usually sells.

> **About linear regression.** The test plan names linear regression as the method. In the system it is one part (30%) of the simple trend, which is used only for products with two to six weeks of sales. The main method is the Random Forest and Gradient Boosting ensemble. The documents should say the same as the system.

8. The last step: the forecast corrects itself

Every method reads a product's level from its recent weeks. When sales are falling, such a forecast runs high; when they are rising, it runs low. In testing, the forecast was above what sold in every fortnight, because the shop's sales fell fortnight after fortnight.

The system tests itself each time it runs (section 10). That test shows how far off it was, most recently, on days it had not seen. Today's forecast is corrected by **half** of that:

```
factor = 1 + 0.5 × (sold ÷ forecast − 1), kept between 0.6 and 1.4
```

Example. In the last fortnight the method forecast 100 and 80 sold. Sold ÷ forecast is 0.8, so the factor is 1 + 0.5 × (0.8 − 1) = 0.9. Today's forecast is multiplied by 0.9.

- **Half, not all**, because one fortnight's miss is partly luck, and correcting all of it would make the forecast swing.
- **Category by category**, because categories do not rise and fall together. A category that sold fewer than 20 items in the test fortnight uses the whole shop's factor.
- **It never looks ahead.** The factor comes only from days already past.

9. From the forecast to a decision

"Daily" below means the 30-day forecast divided by 30. Delivery is taken to be 7 days for every supplier.

What the page shows	How it is worked out
Stock lasts	Stock ÷ daily sales
Reorder now	Stock is at or below the reorder level, or lasts 7 days or less
Reorder soon	Stock lasts 21 days or less
Order by	The run-out date less 7 days (today, if already at the reorder level)
Suggested order	Enough for the next 30 days plus the 7-day wait, and never less than what brings stock back to twice the reorder level; less anything a supplier is already bringing
Safety stock	A buffer for days when sales run above average: 1.65 × how much daily sales vary × √7. The more a product's sales jump around, the larger it is
Reorder point	7 days of expected sales + safety stock
Too much stock	Whatever is left after 3 months of expected sales
Likely to expire	Stock less what is expected to sell before the expiry date
Selling fast / slowly	The top and bottom quarter of products by expected sales
Not selling	Almost nothing expected, and nothing sold in the last 30 days

A worked example

A product is forecast to sell 30 in the next 30 days, so 1 a day. There are 12 in stock. Its reorder level is 10. Its daily sales vary by about 1.

1. **Stock lasts:** 12 ÷ 1 = 12 days.
2. **Status:** 12 days is more than 7 and no more than 21, and 12 is above the reorder level of 10, so it reads **Reorder soon**.
3. **Order by:** it runs out in 12 days; less 7 days for delivery, the order should go in within 5 days.
4. **Suggested order:** 30 for the month + 7 for the wait − 12 in stock = **25**. (Bringing stock back to twice the reorder level would need only 8, so 25 stands.)
5. **Safety stock:** 1.65 × 1 × √7 = 4.4, rounded up to **5**.
6. **Reorder point:** 7 + 5 = **12**. The page would say: recommended reorder point 12, which includes a safety stock of 5 (it is currently set to 10).

10. How its accuracy is measured

The test

1. Hide the most recent 14 days.
2. Forecast them using only the days before.
3. Compare the forecast with what really sold.

The system does this by itself every time the forecast is worked out, and shows the result on the Stock planning page.

Accuracy and error are two different things

The page shows **accuracy**, for the shop total:

> `accuracy = 100 − (how far the forecast total was from the sold total, as a percentage)`

Error is measured product by product. The two can tell very different stories.

> **Example.** Two products. The forecast says 3 and 1. They sell 1 and 3. The total forecast is 4 and the total sold is 4, so accuracy is **100%**. Yet each product was off by 2, so the average error per product is **2 items**. Both statements are true. Always say which one a figure is.

Measure	In plain words
MAE	The average size of the miss, in items, per product per fortnight
RMSE	The same, but large misses count for more
MAPE	The average miss as a percentage of what sold. It cannot be worked out where nothing sold, which here is most of the time, so it is a poor ruler for this shop
WAPE	The total miss as a percentage of the total sold
Bias	Whether the forecast leans one way: forecast total against sold total. Above zero means it runs high

11. The results

The forecast was tested on four fortnights it was not shown, from 16 August to 10 October 2026: 120 products in each, 780 items sold in all.

Before and after the fixes

	Average miss per product (MAE)	Large misses (RMSE)	Leans high by (bias)	Forecast against sold
Before the fixes	1.55 items	1.92	+36.4%	1,064 against 780
Now	**1.40 items**	**1.74**	**+15.6%**	**901 against 780**

Fortnight	Sold	Page accuracy before	Page accuracy now
16 to 29 August	235	63.5%	70.0%
30 August to 12 September	210	57.3%	78.7%
13 to 26 September	175	75.8%	90.9%
27 September to 10 October	160	58.5%	86.1%

Product group	Leaned high by, before	Now
Food	+67.9%	+37.2%

Product group	Leaned high by, before	Now
Health & Wellness	+18.8%	+2.9%
Accessories	+16.1%	+2.8%
Litter & Waste	+37.6%	+15.8%

Against simple rules of thumb

Way of forecasting	Average miss (MAE)	Large misses (RMSE)
The system now	**1.40**	**1.74**
"The average of the last 28 days"	1.58	2.14
"The same as the last 14 days"	1.65	2.42
"Forecast nothing at all"	1.63	2.42

The system is 11% better than the best rule of thumb on the average miss, and 19% better on large misses. Before the fixes it was 2% and 10% better.

> **Read the last row carefully.** Forecasting nothing at all misses by 1.63 items, because a product here sells only one or two items in a fortnight. No method can have a small error per product on sales this thin. The forecast earns its keep on totals: for Health & Wellness and for Accessories it is now within 3% of what sold.

The honest figure to quote

The settings were chosen by a test in which each fortnight was scored with settings picked **without sight of that fortnight** (next section). That result is: **bias +18.9%, average miss 1.45 items**. Quote this one as the tested result. The +15.6% above is the system as it now stands, on fortnights that also helped choose its final settings, so it is slightly flattering.

12. How the settings were chosen

The forecast has a few settings: how far back the model reads, how fast older days fade, how much of the final figure is the model and how much the simple trend, and what kind of last-step correction to apply. In all, 1,620 combinations were tried.

The danger in choosing settings is choosing the ones that happen to fit the test. Two things guard against it:

- **The rule was fixed first**, before any result was looked at: keep the combinations whose forecast total is within 10% of what sold, and among those take the one with the smallest error per product.
- **Each test fortnight was scored with settings chosen only from earlier fortnights.** It is like a mock exam marked with an answer key that was sealed before the exam.

What the rule chose, and what the system is now set to:

Setting	Before	Now
How far back the model reads	365 days	120 days
How older days count	Every day the same	A day's weight halves every 60 days
The model's share of the final figure	Half (the simple trend had the other half)	All of it, for a product with 42 days or more
Last step	None	Corrected by half of its own last miss, by category

One thing the test found: neither half of the old half-and-half mix was the cause of the forecast running high. Sales fell every fortnight from mid-July (362, 335, 299, 235, 210, 175, 160 items), and both halves read the level from recent weeks, so both ran high. The model had the smaller error per product in every period tested.

13. What the forecast cannot do

Say these whenever the forecast is presented. Stating them is a strength, not a weakness. The figures for made-up products below are the forecast before its last-step correction, which would then multiply each by its category's factor.

Limit	What it means in practice
Per product, the error is as large as the sales	Use the forecast for a category or the shop. Treat one product's figure as a rough guide
It still runs high	By about 19% on fortnights it had not seen. Food is the worst group
Its settings were chosen while sales were only falling	It has never been tested in a period of rising sales, because the shop has none on record. The test should be run again when a few more months exist
A lasting rise is recognised slowly	A made-up product that went from 1 sale every 5 days to 1 a day was half recognised after about six weeks. Check fast-rising products by eye
A product in sharp decline can be kept too high	This was finding L4 in the QA report and it is not fixed. A rule for it was tried, but it never acted on any real product, so it could not be tested here
An empty shelf looks like no demand	The system reads sales, not what customers wanted. A product that was out of stock for two weeks looks like a product that sells less
Seasons and holidays	225 days cannot show a yearly season such as the weeks before Christmas. Holy Week, promotions, price changes and a supplier running short are not things it is told
Confidence reads "low" for every product	That is the honest reading: no product sells often enough for the system to call its forecast reliable

Limit	What it means in practice
The sales history itself	The QA report found the sales on record did not go through the till, and that their level falls to about a quarter in June. Every figure here describes how the method does on this history. Be ready to say how the history was made
Fixed assumptions	Delivery is taken to be 7 days for every supplier. Neither that nor the safety buffer is learned from data

14. Questions you may be asked

Why Random Forest and Gradient Boosting, and not linear regression?

A straight line cannot follow weekends, pay-day weeks or a product that sells now and then. On the four test fortnights the simple trend, which contains the linear regression, missed by 1.58 items per product; the two models missed by 1.47, and 1.40 with the last step.

The page says 86% accuracy. Is the forecast 86% right?

For the shop total in that fortnight, yes: 182 forecast against 160 sold. For a single product, no. A single product's forecast misses by about 1.4 items on average, which is about as much as it sells. Section 10 explains the difference.

Why does every product say "low" confidence?

Because no product sold on more than 13 of the last 56 days. The system used to call 14 products "high": they were the ones that had stopped selling, and "no sales" was being read as "perfectly steady sales". That is fixed.

Why were 22 days in May left out?

In those days sales were recorded without their items, or not at all. They say nothing about what customers wanted, and counting them as "nothing sold" taught the forecast a false lesson. Nothing was deleted; the days are only skipped when the history is read.

Could one wrong entry ruin the forecast?

It could before. A single day of 10,000 in one product raised every other product's forecast by 130%. Now a day counts for no more than four times the product's typical selling day, and in the same test the freak day adds 0.3 of a percentage point to what that product does without it.

Is the forecast biased?

Yes, it runs high: by about 19% on fortnights it had not seen, down from 36%. The cause is that sales have been falling, and any method that reads recent weeks lags behind a fall. The last step corrects half of the most recent miss.

How do you know the settings were not just fitted to the test?

The rule for choosing was written down before any result was seen, and each test fortnight was scored with settings chosen only from fortnights before it.

What happens with a brand-new product?

For its first two weeks its sales so far are spread over 14 days, so one good first day cannot trigger a large order. From two to six weeks a simple trend is used. After six weeks the models take over.

What happens if the machine-learning program stops working?

Every product falls back to the simple trend, the page says so, and the server log records why. The forecast does not fail.

15. Words used in this part

Word	Meaning
Forecast	An estimate of future sales, worked out from past sales
Model	A program that has learned a pattern from past data and applies it to new cases
Decision tree	A chain of yes-or-no questions ending in an estimate
Random Forest	Many decision trees, each grown on a different sample; their answers are averaged
Gradient Boosting	Small trees built one after another, each correcting the ones before
Ensemble	Two or more models whose answers are combined
Held-out fortnight	Two weeks of sales hidden from the forecast, then used to check it
Walk-forward test	Testing one period at a time, always using only what was known before it
Outlier	A value far from the usual, such as one day of 10,000
Bias	A steady lean to one side: a forecast that is usually too high, or usually too low
Lead time	The days between placing an order and receiving it (7 in this system)
Safety stock	Extra stock kept in case sales run above the forecast
Reorder point	The stock level at which an order should be placed
SKU	The shop's own code for a product